

ASCII MASTER FLOW – PANEL 1/12: Human-rating pyramid

LAUNCH RELIABILITY -> necessary starting condition
MARGINS AND REDUNDANCY -> tolerate selected failures
HEALTH MONITORING -> detect unsafe vehicle conditions
ABORT MODES -> protect crew during ascent
HUMAN RATING -> system qualification, not a success count

ASCII MASTER FLOW – PANEL 2/12: Gaganyaan module map

CREW MODULE -> habitable and re-entry capable

SERVICE MODULE -> propulsion, power and support

ORBITAL MODULE -> Crew Module plus Service Module

LAUNCH VEHICLE -> ascent platform, separate from spacecraft

RULE -> module, spacecraft and launcher are different layers

ASCII MASTER FLOW – PANEL 3/12: Crew-survival chain

ECLSS -> breathable and thermally controlled cabin

CREW ESCAPE -> separation from a failing launcher

RE-ENTRY -> controlled atmospheric return

PARACHUTES -> staged deceleration

RECOVERY -> splashdown, extraction and medical support

ASCII MASTER FLOW – PANEL 4/12: Qualification ladder

COMPONENT / GROUND TEST -> verify subsystem behaviour

AIR-DROP TEST -> parachute and recovery sequence

TEST-VEHICLE ABORT -> escape under a selected flight condition

UNCREWED ORBITAL FLIGHT -> integrated mission without crew

CREWED FLIGHT -> later accepted human risk

ASCII MASTER FLOW – PANEL 5/12: International-exposure firewall

AXIOM-4 -> international ISS mission experience

FOREIGN LAUNCHER / SPACECRAFT -> external transport systems

INDIAN EXPERIMENTS -> operational and scientific exposure

GAGANYAAN -> indigenous launch, module and recovery objective

RULE -> experience complements but does not certify capability

ASCII MASTER FLOW – PANEL 6/12: Mission-status ladder

ANNOUNCED / PROPOSED -> intent

APPROVED -> authority and programme sanction

UNDER DEVELOPMENT / TESTED -> work and evidence rung

LAUNCHED / INSERTED / OPERATIONAL -> distinct flight milestones

COMPLETED / PARTIAL / FAILED / CONCLUDED -> outcome vocabulary

ASCII MASTER FLOW – PANEL 7/12: Lunar capability ladder

CHANDRAYAAN-1 -> orbital science and lunar-water evidence

CHANDRAYAAN-2 -> orbiter success, lander setback

CHANDRAYAAN-3 -> soft landing and surface operations

CHANDRAYAAN-4 -> approved sample-return technology mission

RULE -> landing does not prove ascent, docking or Earth return

ASCII MASTER FLOW – PANEL 8/12: Solar-observatory geometry

SUN-EARTH SYSTEM -> defines Lagrange-point geometry
L1 REGION -> continuous solar-view advantage
HALO ORBIT -> three-dimensional path around L1
STATION KEEPING -> periodic correction and propellant use
ADITYA-L1 -> solar observatory, not a landing mission

ASCII MASTER FLOW – PANEL 9/12: Planetary-status matrix

MARS ORBITER MISSION -> historic mission, now concluded

VENUS ORBITER MISSION -> approved future mission in owner

LUNAR SAMPLE RETURN -> ascent and return architecture

SOLAR OBSERVATORY -> remote sensing from L1 halo orbit

RULE -> destination and status both control the description

ASCII MASTER FLOW – PANEL 10/12: Parallel capability ladders

HUMAN -> abort, life support, uncrewed, crewed, station

LUNAR -> orbit, soft land, rove, sample return

SOLAR -> transit, halo insertion, station keeping, observation

PLANETARY -> cruise, insertion, science operations

RULE -> one ladder cannot certify another

MUST REMEMBER: Human spaceflight requires a human-rated launcher, crew and service...

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ASCII MASTER FLOW – PANEL 11/12: Planetary-defence rail

DETECTION -> discover potentially hazardous objects

CHARACTERISATION -> refine orbit, size and composition

WARNING TIME -> determines response feasibility

DEFLECTION -> kinetic impact is one demonstrated concept

RESIDUAL RISK -> one test is not a complete shield

CLOSE DISTINCTION: Launch-vehicle test is not crew-system qualification, abort test is...

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ASCII MASTER FLOW – PANEL 12/12: Mission answer spine

CLASSIFY -> human lunar solar planetary or defence mission

TRACE -> minimum technical and safety chain

NAME -> one verified achievement or approved next rung

SEPARATE -> test launch operation completion and outcome

VERIFY -> dated crew schedule payload finding and status source

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